Introduction

The program is designed to manage information about a set of people stored in two separate text files. The first file contains various information about each person, including their ID, exam scores, course details, number of classes for instructors, etc. The second file maps each person's ID to their name, surname, role, and department. The program consists of utility functions for processing data from these files and a query interface allowing users to perform various operations like calculating GPA, printing pass or fail status, finding the number of classes for instructors, etc.

Part 1: Utility Functions

1. `first_initial` (FILE * id_fp, int id)

- This function scans the ID file and returns the initial letter of the first name of the person with the given ID.

2. `last_initial` (FILE * id_fp, int id)

- Similar to the `first_initial` function, but it returns the initial letter of the person's last name.

3. `get_id_fi` (FILE * id_fp, char first_initial)

- Returns the ID of a person given the first initial of their name.

4. `get_id_li` (FILE * id_fp, char last_initial)

- Returns the ID of a person given the last initial of their name.

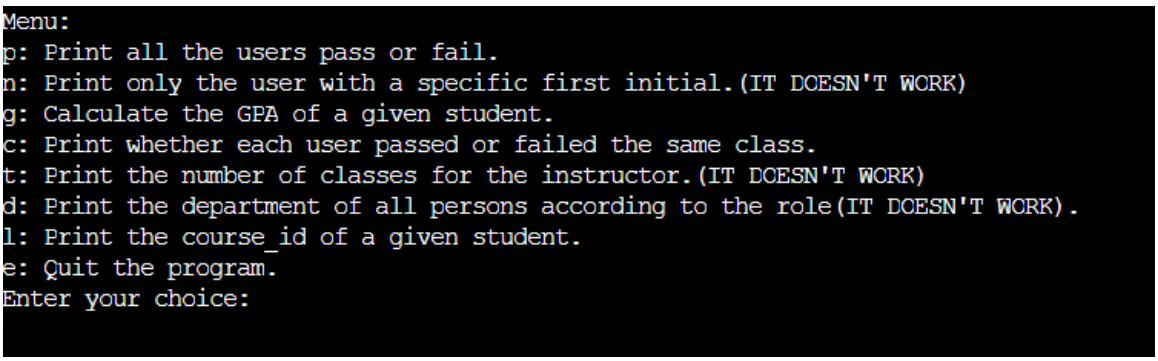
5. `average_grade` (FILE * info_fp, int id)

- Calculates and returns the GPA of a person defined by the given ID.

Part 2: Query

The program provides a menu-based interface with the following options:

MENU:



'p': Print all users pass or fail.

- Iterates through all students and calculates their letter grades based on the provided exam scores.

- Displays whether each student has passed or failed.

```
Printing pass or fail for each user:
ID: 220015001, Result: P
ID: 220015002, Result: P
ID: 220015003, Result: P
ID: 220015004, Result: P
ID: 220015005, Result: P
ID: 220015006, Result: P
ID: 220015007, Result: P
ID: 220015008, Result: F
ID: 220015009, Result: P
ID: 220015010, Result: F
ID: 220015011, Result: F
ID: 220015012, Result: F
ID: 220015013, Result: F
ID: 220015014, Result: F
ID: 220015015, Result: P
ID: 220015016, Result: P
ID: 220015017, Result: F
ID: 220015018, Result: P
ID: 220015019, Result: P
ID: 220015020, Result: F
ID: 220015021, Result: F
ID: 220015022, Result: P
ID: 220015023, Result: F
ID: 220015024, Result: F
ID: 220015025, Result: P
```

'n': Print only the user with a specific first initial.

- Asks the user to enter a letter for the initial (first name).

- Displays pass/fail status for students matching the entered initial.

'g': Calculate the GPA of a given student.

- Asks the user for the ID of the student.

- Calculates the GPA of the student based on their grades.

```
Calculating GPA for student with ID 220015001:
GPA: F
```

'c': Print whether each user passed or failed the same class.

- Asks the user to enter a course ID.

- Calculates the average grade for students in the specified course and displays pass/fail status.

't': Print the number of classes for the instructor.

- Asks the user for the instructor's ID.

- Finds the number of classes taught by the instructor.

'd': Print the department of all persons according to the role.

- Asks the user to enter a letter for the initial (role).

- Finds all persons' IDs and departments matching the role and displays them.

'l': Print the course ID of a given student.

- Asks the user for the student's ID.

- Finds and displays the course IDs associated with the student.

```
Printing the course ID of the given student (Student ID: 220015001):
Course ID: 68
```

'e': Quit the program.

```
Menu:
p: Print all the users pass or fail.
n: Print only the user with a specific first initial.(IT DOESN'T WORK)
g: Calculate the GPA of a given student.
c: Print whether each user passed or failed the same class.
t: Print the number of classes for the instructor.(IT DOESN'T WORK)
d: Print the department of all persons according to the role(IT DOESN'T WORK).
l: Print the course_id of a given student.
e: Quit the program.
Enter your choice: e

Exiting...
```

The program allows users to continue selecting options until they choose to exit.

Conclusion

The program provides a user-friendly interface for querying and managing information about people stored in text files. It includes utility functions for processing data and various query options to retrieve specific information based on user input. By following the provided instructions and functionalities, users can efficiently interact with the system to obtain relevant data about students, instructors, courses, and departments.